

{ Blue point used for ^{the} answers marked in the paper }

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MTH634 - TOPOLOGY

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5:00 pm

MCQ'S

① Let $X = \{1, 2, 3\}$, then $P(X) =$

* $\{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, X\}$

② Let $X = \mathbb{R}^2$ with usual topology and

$A = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < 1\}$. Closure of A is

* $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$

③ For a subset A of a topological space X , Let $\bar{A}$ denotes closure of A and A' denotes the derived set of A , then A is closed if and only if

* $\bar{A} = A$

④ Let $X = \{a, b, c, d\}$ and $\mathcal{T} = \{\emptyset, \{a\}, \{a, b\}, \{a, c, d\}, X\}$ be $\mathcal{T}$ on X . The interior point of

$A = \{a, b, c\}$ is

* c

* d

* a

* none

⑤ Let $X = \{a, b, c, d\}$ with $\mathcal{T} = \{\emptyset, \{a\}, \{a, b\}, \{b\}, \{a, c, d\}, X\}$. The basis for $\mathcal{T}$ is

* $\{\{a\}, \{b\}, \{a, c, d\}\}$

⑥ Let X be any topological space and Y be Indiscrete Space then any $f: X \rightarrow Y$ is

* continuous

* not continuous

* none

⑦ If a function $f: X \rightarrow Y$ is continuous at every point of X , then f is

* Continuous * not continuous * none

⑧ A map $f: (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$ is called a closed map iff image of each closed subset of X closed in Y

⑨ Which one following of the statement is False?

* Boundedness is a topological property.

⑩ Which one of the following statement is False?

* Co-finite $\mathcal{T}$ on a Real Line $\mathbb{R}$ is 1st countable

⑪ Every subspace of second countable is second countable

* True * False

⑫ For $x, y \in \mathbb{R}$, we define the usual metric on $\mathbb{R}$ as $d(x, y) = |x - y|$

* $d(x, y) = |x - y|$

⑬ Let $B_1(x, r_1)$ and $B_2(x, r_2)$ be two open balls centered at x with radii r_1 and r_2 respectively. Then the following statement is true.

* $B_1(x, r_1) = B_2(x, r_2)$

* either $B_1(x, r_2) \subset B_2(x, r_2)$ or $B_1(x, r_1) \subset B_2(x, r_1)$

* none

⑭ Let $X = \mathbb{R}$ with usual topology which / is not an open cover of $\mathbb{R}$
 $\mathcal{U} = \{(-n, n) \mid n \in \mathbb{N}\}$

* ~~$\mathcal{U}$ is a cover of $\mathbb{R}$~~

* ~~$W = \{(n, n+1) \mid n \in \mathbb{Z}\}$~~

* ~~$V = \{(n, n+2) \mid n \in \mathbb{Z}\}$~~

* $W = \{[n, n+1] \mid n \in \mathbb{Z}\}$

* none

⑮ Which of the following statement is True?

* Every separable space is second countable.

* Every second countable space is separable.

* $\mathbb{R}$ with Lower Limit topology is second countable.

⑯ Which of the following statement is True?

* A separable metric space is second countable.

* Separable metric space is first countable.

* Every metric space is second countable

⑰ A topological space is a T_1 -space if and only if its each finite subset is

* a perfect set

* a closed set

* an open set

* none

18) A finite set has — is a T_1 space.

- * No isolated points
- * No limit points
- * Infinite limit point

19) Every subspace of T_3 -space is a

- * T_1 -space
- * T_2 -space
- * T_3 -space

20) Which of the following statement is false?

- * Every metric space is second countable
- * Every metric space is Hausdorff space.
- * Every metric space is normal space.

21) Every discrete space is not a regular space

- * True
- * False

22) A set X with co-finite topology is compact.

- * False
- * True

23) Every closed subset of a compact space is compact

- * True
- * False

24) Consider R with usual topology / (R, τ_u) , there exist no homeomorphism between an open interval of R and closed interval of R

- * True
- * False

25) Consider R with usual topology. Then $A = (0, 1) \cup [3, 5]$ is disconnected subset of R

- * True
- * False

(26) The image of ^{connected} countable space under a continuous map is not connected.
* True * False

(27) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as $f(x) = x+1$. Fixed point of f is
* 0 * 1 * 2 * no fixed point

(28) The union of a collection of connected subspaces of $(X, \mathcal{T})$, that having a point common, is connected.
* True * False

(29) A connected compact C of a topological space X is not open in general.
* True * False

(30) If $(X, \mathcal{T})$ be a path connected topological space then $(X, \mathcal{T})$ is connected.
* True * False

SUBJECTIVE PORTION

Q1) Let $X = \{1, 2, 3, 4\}$. Write the basis for the discrete ~~for~~ ~~the~~ on X .

Q2) Define separated sets.

Q3) Let $(X, \mathcal{T})$ and $(Y, \mathcal{T}')$ be topological spaces. Prove that a function $f: X \rightarrow Y$ is continuous ~~if~~ the inverse of each member of a base β for Y is an open subset of X .

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Q4) Prove that Every subspace of a second countable space is second countable.

Q5) Let A be connected subset of a topological space $(X, \mathcal{T})$ & $A \subset B \subset \bar{A}$. Show B is connected.

Q6) Let $\mathcal{T}$ be cofinite topology on any set X . Show that $(X, \mathcal{T})$ is separable. i.e. contains a countable dense subset.

Q7) Define Urysohn's Lemma.

Q8) Name two topological properties of topological space $(X, \mathcal{T})$.

Q9) Consider $\mathcal{T} = \{\emptyset, \{a\}, \{b\}, \{a, b\}, X\}$ on the set $X = \{a, b, c\}$.

Show that $(X, \mathcal{T})$ is a normal space.

Q10) Consider $X = \{a, b, c, d, e\}$ with

$$\mathcal{T} = \{\emptyset, \{a, b\}, \{d, e\}, \{a, b, c\}, \{a, b, d, e\}, X\}$$

Then X is a disconnected space.

Q11) Show that every second countable space is First countable.